

Rupak Neupane

📍 Bhaktapur, Nepal ✉ neupanerupak7@gmail.com ☎ +977-9865442388 🌐 rupakneupane.com.np
in rupakneupane007 📱 RupakNeupane

Education

Khwopa College of Engineering, Tribhuvan University 31 April 2021 - 15 May 2025
Bachelor of Computer Engineering

- **Entrance Top Ranker Scholarship 2077:** Secured a full scholarship for undergraduate study awarded to the top-ranking applicant from each program in the IOE Tribhuvan University entrance examination, among candidates applying to Khwopa College of Engineering.
- **Core Coursework:** Computer Programming, Theory of Computation, Data Structure and Algorithm, Data Mining, Big Data Technologies, Computer Network, Operating System, DBMS, AI.

Overall Percentage: 80.32%

Budhanilkantha School 17 July 2018 - 10 June 2020
GCE A Levels

- **Coursework:** Physics - A*, Chemistry - A*, Computer Science - A*, Mathematics - A*, English Literature - d and General Paper - d

Work Experience

Artificial Intelligence Developer 21 October 2025 - 05 January 2026
Global IME Bank

- **Intelligent Invoice Extraction:** Engineered a multilingual (Nepali/English) extraction system using state-of-the-art Vision-Language Models (VLM); implemented Reinforcement Fine-Tuning to adapt models for complex banking contracts, achieving significant gains in field-level extraction accuracy.
- **Document Query System (DQS):** Architected an internal RAG (Retrieval-Augmented Generation) system to democratize access to institutional knowledge, allowing employees to query complex internal documents via natural language.

Projects

Word-Embedding in Nepali Language (Sabdhamanthan)

- Developed an AI model using the BERT architecture for context-dependent word embedding that understands the Nepali context and can be used for many downstream tasks using this understanding of the Nepali language.

Facial Keypoint Detection

- Developed a deep learning-based facial landmark detection system using a modified VGG-16 architecture to accurately predict 68 facial keypoints for computer vision applications.

25 Bird Image Classification

- Developed a deep learning model using a custom ResNet9 architecture to classify 25 bird species from images, focusing on dataset pre-processing, model training, and classification.

Publications

A Solution to The Face DeepFake Detection Challenge April 2025
Rupak Neupane, Srijan Gyawali, Sarjyant Shrestha, and Manish Pyakurel
Proceedings of the 16th IOE Graduate Conference, Vol. 16, pp. 2023-2028
Institute of Engineering, Tribhuvan University, Nepal 📄

Automating Document Workflows with ResNet-50 and Template-Based OCR July 2025
Srijan Gyawali, Rupak Neupane, Sarjyant Shrestha, and Manish Pyakurel
Journal of Engineering Issues and Solutions, Vol. 4, Issue 1, pp. 478-485
DOI: 10.3126/joeis.v4i1.81610 📄

Skills

Technical Skills

- Python, Pytorch, Tensorflow, Keras, NLP, Computer Vision, Cloud Computing (AWS, GCP), C++, C, SQL, LaTeX

Soft Skills

- Problem Solving, Leadership, Teamwork, Critical Thinking

Achievements

1st Runner-up, Global IME AI/ML Hackathon

02 May 2025

Global IME Bank

- Developed a document classifier and text extraction system for bank-related documents

Mentor at a Hackathon "HackTheCircle"

07 December 2024

Khwopa College of Engineering

- Mentored the participants about using AI/ML in their projects at hackathon

Lead Instruction at LaTeX Workshop

28 June 2024

Collabyte — Khwopa College of Engineering

- Conducted a LaTeX Workshop in which we taught from basics and dived into advanced concepts in LaTeX

Best Technical Implementation Award, Software Competition

17 December 2024

Khwopa College of Engineering

- Developed an event management system for effectively managing the booking system of various halls and rooms around the college.

Top 50 qualifiers at "Kathmandu Metro Idea and Innovations 2080 (KMII 2080)"

Kathmandu Metropolitan

- Went through various rounds of selection by proposal submission, idea pitching, and project showcase to be qualified among all the participants around the nation.

5th Annual Nepal AI School (ANAIS) 2024

27 December 2024 - 06

Nepal Applied Mathematics and Informatics Institute of Research (NAAMII)

January 2025

- Completed a 10-day intensive AI program covering Geometric Deep Learning, LLMs, and Reinforcement Learning, including a hands-on hackathon project.

Certifications

- Microdegree™ in Artificial Intelligence 2025 (Fusemachines) [↗](#)
- Future AWS AI Scientist (Udacity) [↗](#)
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional (Oracle) [↗](#)
- AI Fundamentals (DataCamp) [↗](#)
- Data Literacy (DataCamp) [↗](#)
- AI for Everyone (Coursera) [↗](#)
- Artificial Intelligence Job Simulation (Cognizant (Forage)) [↗](#)
- Introducing Generative AI with AWS (Udacity) [↗](#)
- AWS Educate Machine Learning Foundations (Amazon Web Services) [↗](#)
- AWS Educate Introduction to Generative AI (Amazon Web Services) [↗](#)
- Artificial Intelligence Fundamentals (IBM SkillsBuild) [↗](#)
- Intermediate to Machine Learning (Kaggle) [↗](#)
- Intro to Deep Learning (Kaggle) [↗](#)
- Computer Vision (Kaggle) [↗](#)